Course preparation

1. Install R & RStudio:

Download and install `R` (http://cran.r-project.org/) and `RStudio` (https://docs.posit.co/ide/user/download.html). If you need help, you find useful instructions here: https://rstudio-education.github.io/hopr/starting.html

3. Install and load packages:

Open RStudio, select an R version to work with, open the directory you want to work in (with the `input_data` folder in it) and try to execute this code line in the console:

```
install.packages("ggplot2")
```

If you are questioned of you want to install the packages from source or update other packages, you can accept. If you face problems, try again answering `no`.

Optional, if that doesn't work: If you get error messages that describe that you can't install the packages, the problem is likely that R has no writing access in the directory of your package library.

```
.libPaths() # This shows you the currently used directory (the first entry):
```

```
[1] "C:/R_libraries"
[2] "C:/Users/Thomas/AppData/Local/R/win-library/4.6"
[3] "C:/Program Files/R/R-4.6.1/library"
```

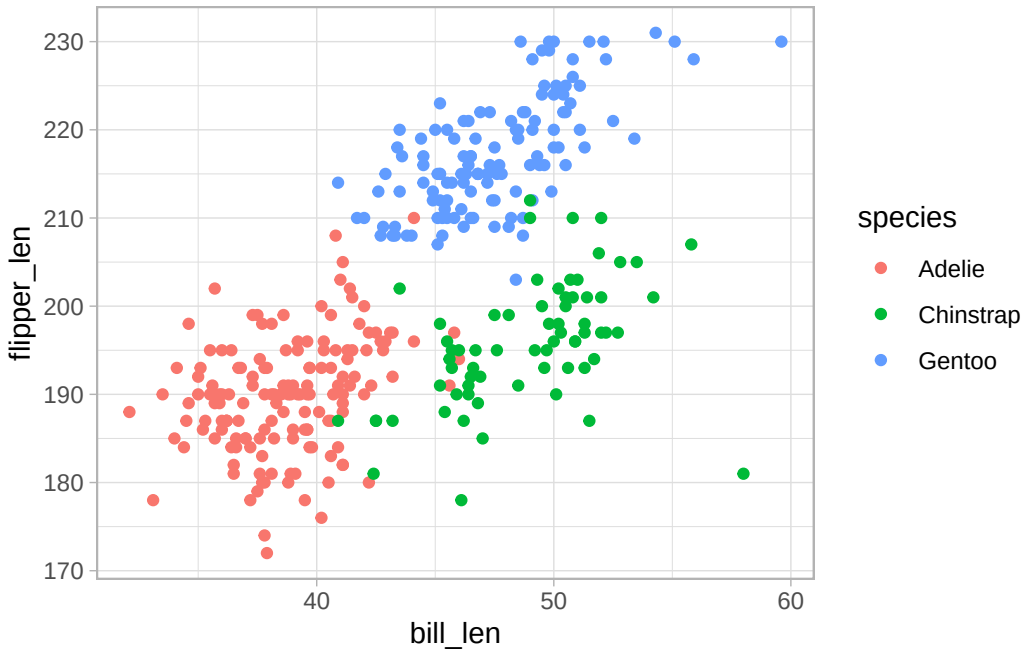
To solve the problem, we generate and use another directory for our libraries:

```
user_lib <- Sys.getenv("R_LIBS_USER") # find personal standard library directory
dir.create(user_lib, recursive = TRUE, showWarnings = FALSE) # Create directory
.libPaths(c(user_lib, .libPaths())) # Activate it
.libPaths() # check if it worked
```

The `.libPaths()` call should show the new directory as first entry. Restart `RStudio` and try to install the package again, using the `install.packages()` line above.

Finally, execute this code to see if the installed packages work:

```r
library(ggplot2); data(penguins)
ggplot(penguins, aes(bill_len, flipper_len, color = species)) +
    theme_light() +
    geom_point(na.rm = T)
```



1. Download DepMap datasets:

Please visit the `DepMap Portal` (https://depmap.org/portal/) page and then to `Downloads` -> `Enter DepMap` -> `All Data`. We need the following files from version `DepMap Public 26Q3`:

Data	Filename
Model metadata	Model.csv
Chronos scores	CRIPSRGeneEffect.csv
Gene Expression	OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv
Mutation status	OmicsSomaticMutations.csv

Download these files and place them in a folder named `input_data`. If this worked, you are ready to go!

If you have questions, feel free to contact me:

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